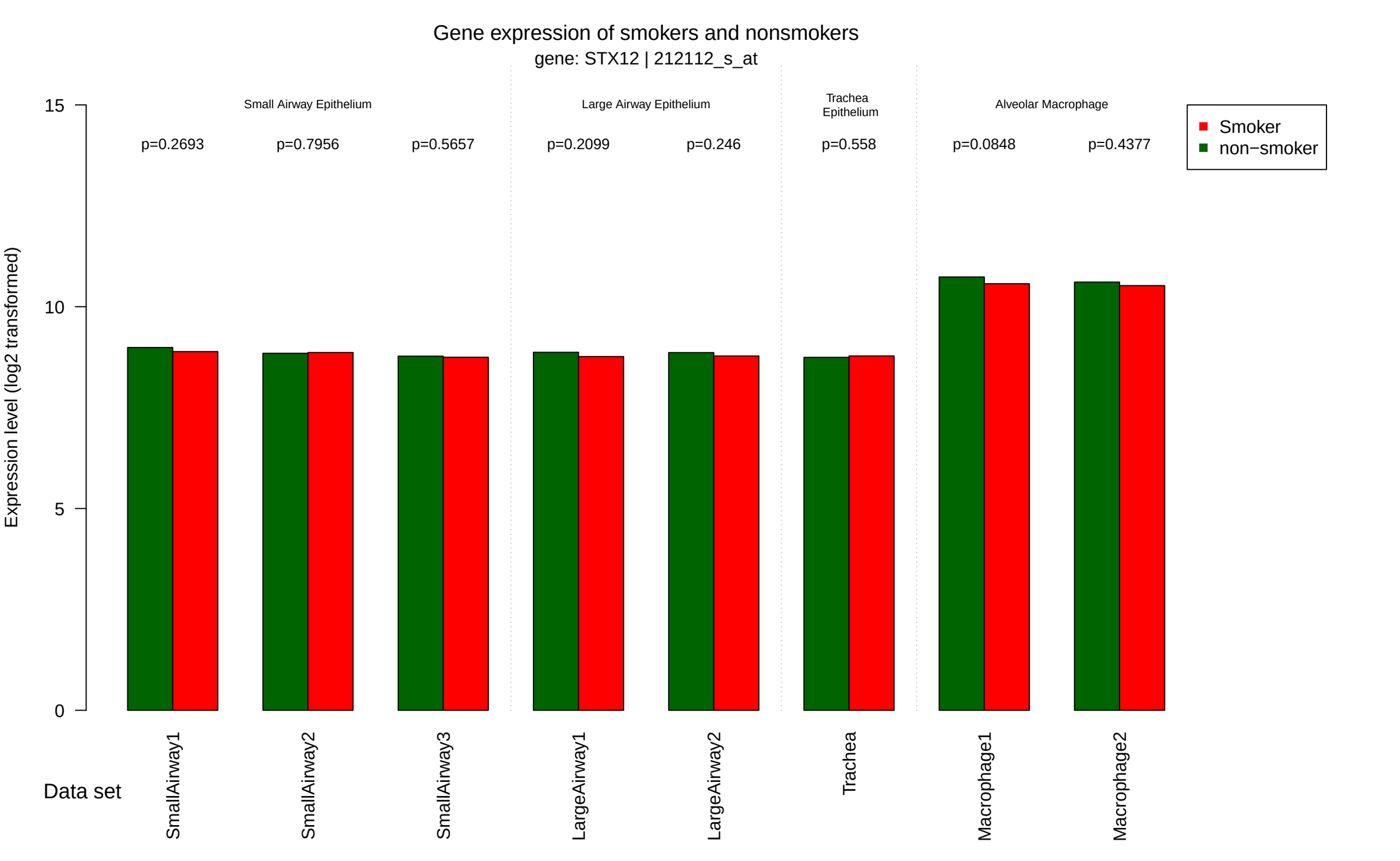
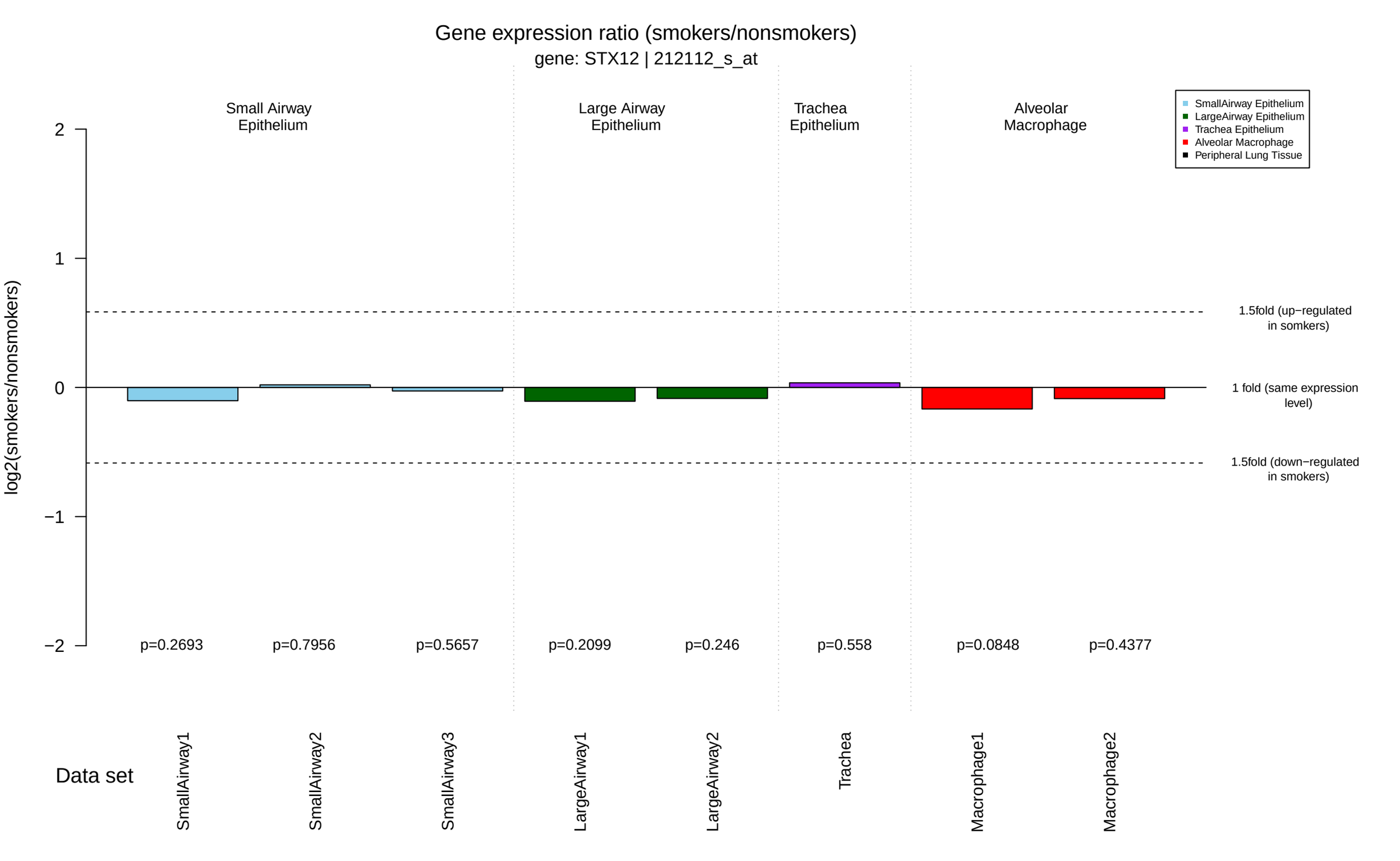
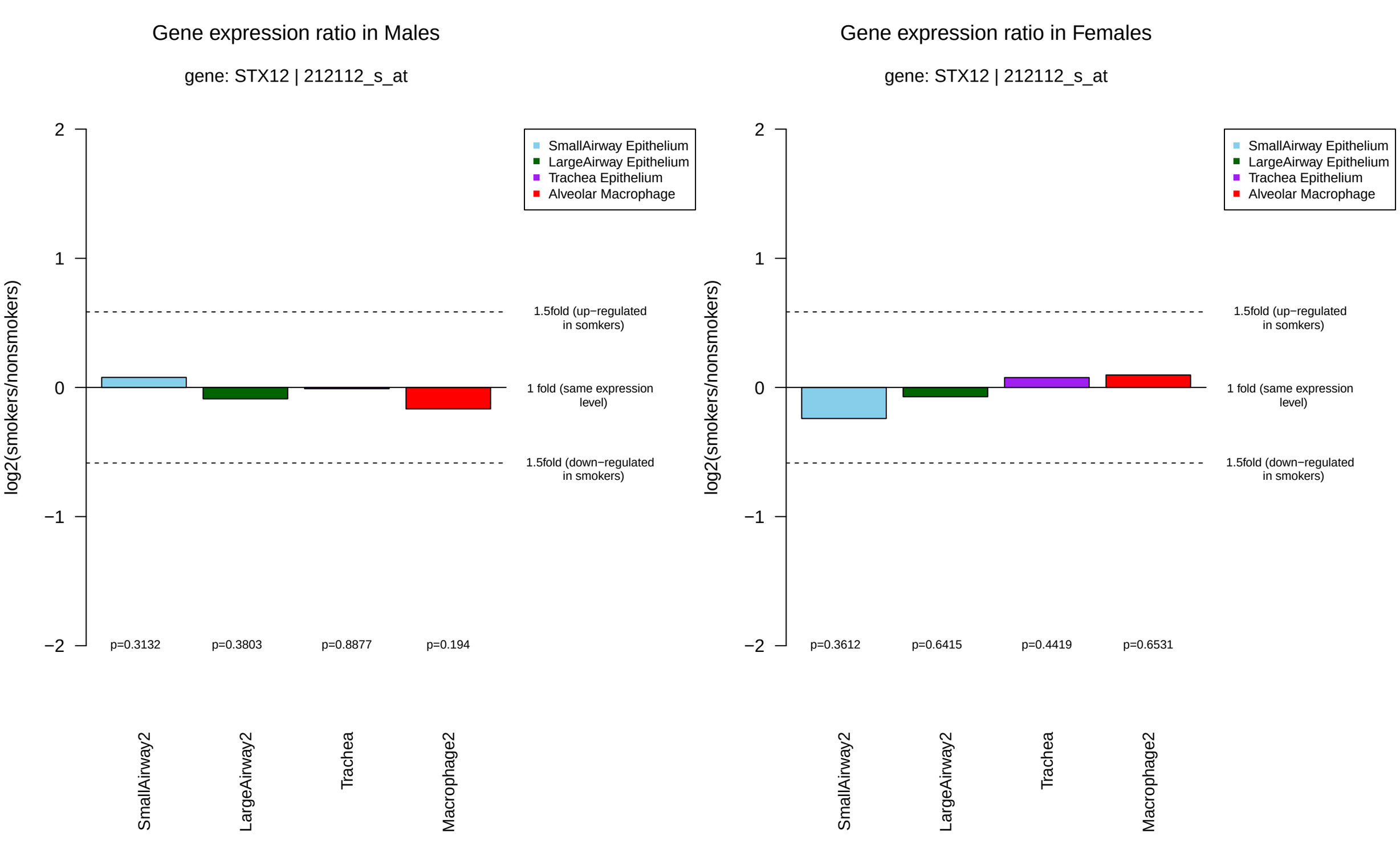
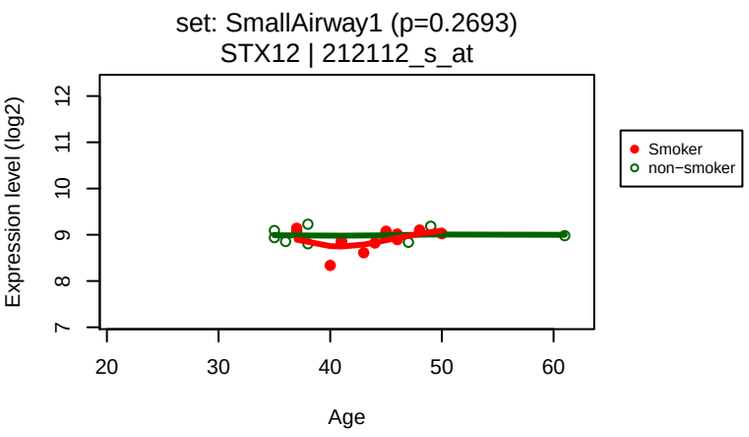
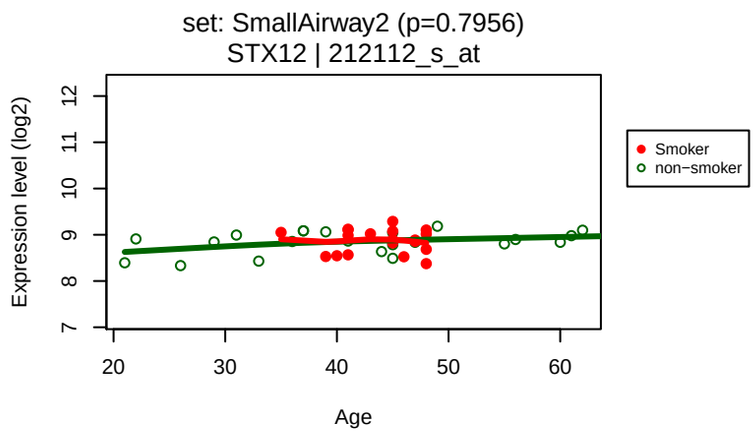
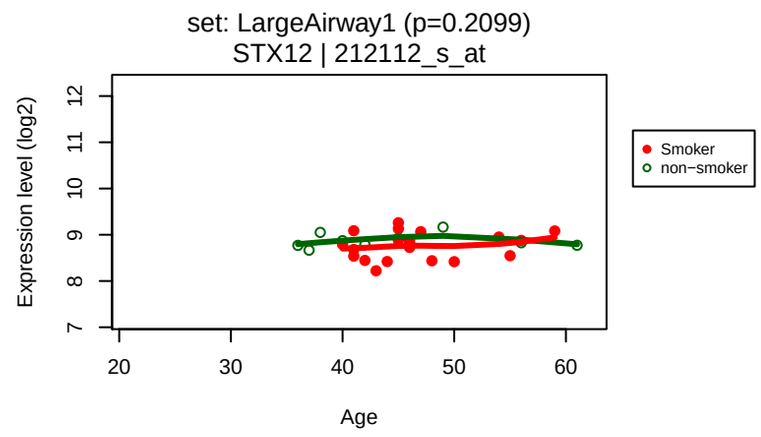

Gene expression of smokers and nonsmokers
gene: STX12 | 212112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2693
p=0.7956
p=0.5657
p=0.2099
p=0.246
p=0.558
p=0.0848
p=0.4377
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

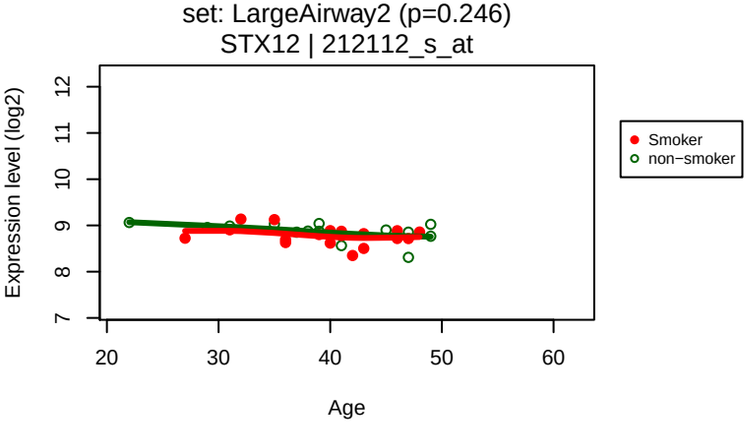

Gene expression ratio (smokers/nonsmokers)
gene: STX12 | 212112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2693
p=0.7956
p=0.5657
p=0.2099
p=0.246
p=0.558
p=0.0848
p=0.4377
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

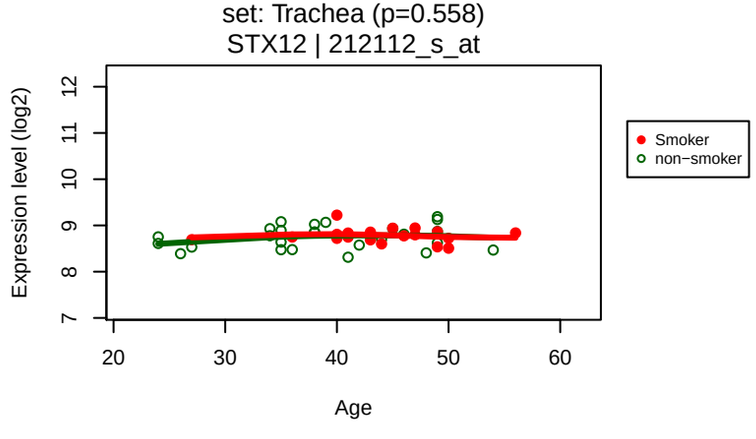

Gene expression ratio in Males
gene: STX12 | 212112_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3132
p=0.3803
p=0.8877
p=0.194
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STX12 | 212112_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3612
p=0.6415
p=0.4419
p=0.6531
SmallAirway2
LargeAirway2
Trachea
Macrophage2

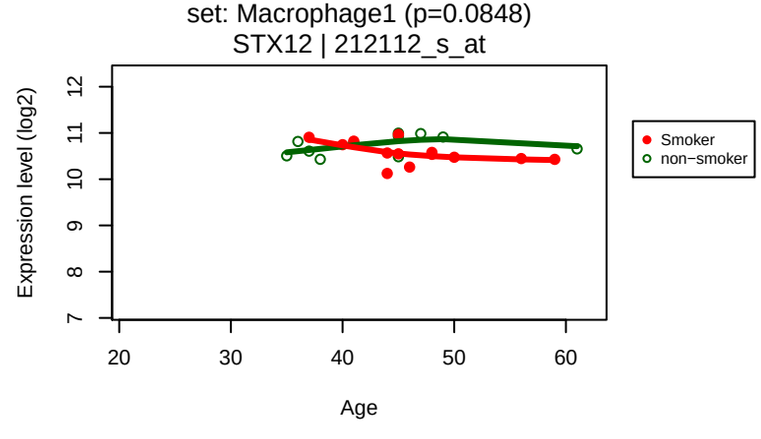

set: SmallAirway1 (p=0.2693)
STX12 | 212112_s_at
Expression level (log2)
Age
Smoker
non−smoker

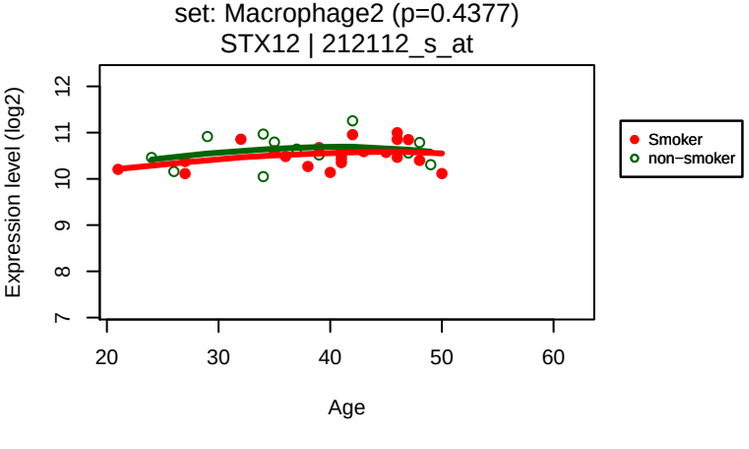

set: SmallAirway2 (p=0.7956)
STX12 | 212112_s_at
Expression level (log2)
Age
Smoker
non−smoker

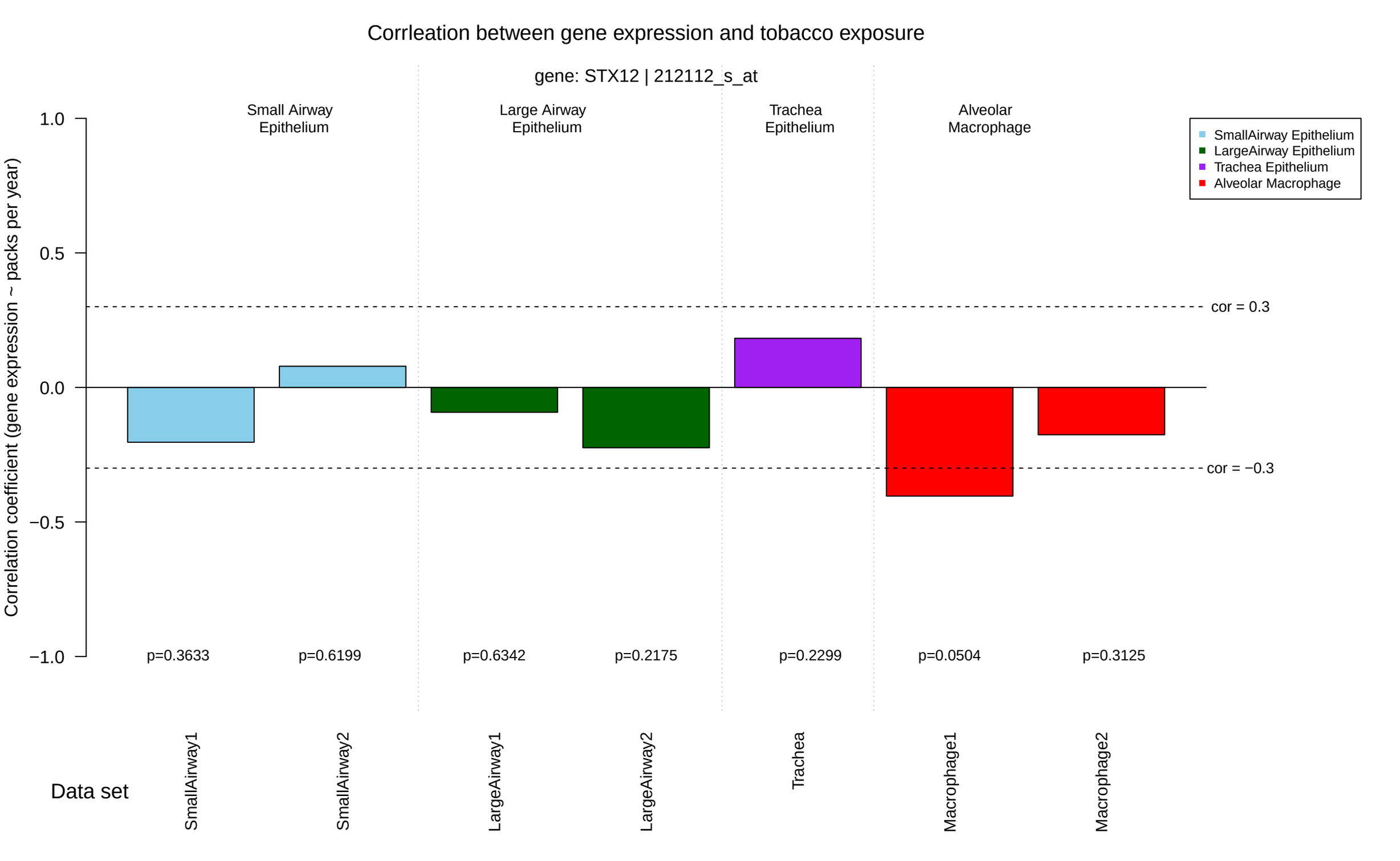

set: LargeAirway1 (p=0.2099)
STX12 | 212112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.246)
STX12 | 212112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.558)
STX12 | 212112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0848)
STX12 | 212112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4377)
STX12 | 212112_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STX12 | 212112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3633
p=0.6199
p=0.6342
p=0.2175
p=0.2299
p=0.0504
p=0.3125
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2